

RSE-Supported Scientific Software Standardization

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- Technical debt → Standardization → Interface
- Steps
 - ① Should you standardize?
 - ② Backward compatibility?
 - ③ Learn necessary domain knowledge
 - ④ Document current design architecture
 - ⑤ Abstraction to isolate interface
 - ⑥ Implementation
 - ⑦ Verification
 - ⑧ Migration plan
- Guidelines
 - Question-driven learning
 - Information hiding
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- Case study: Eelbrain-BIDS

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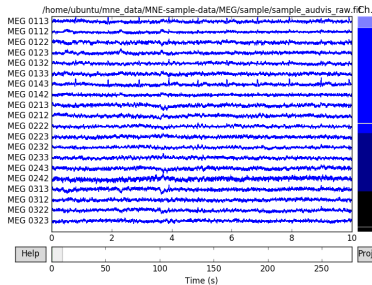
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- Magnetoencephalography (MEG)
- Electroencephalography (EEG)
- Stimulus, then record a multi-channel time series



MEG device and the recorded multi-channel data

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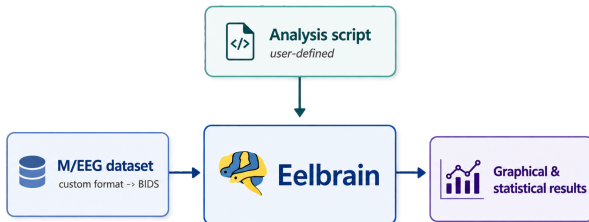
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- Open-source Python library for M/EEG data analysis¹
- BIDS (Brain Imaging Data Structure) is a community standard for organizing and sharing brain data
- The Eelbrain-BIDS project upgraded the input dataset format of Eelbrain from a custom format to BIDS



¹<https://github.com/Eelbrain/Eelbrain>

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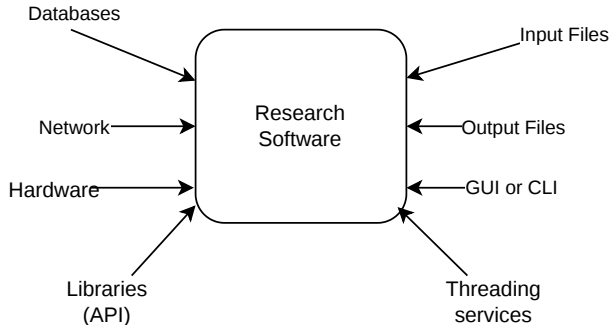
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Technical debt² → Standardization → Interface standards



²Term first used by [Cunningham \(1992\)](#)

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Pros

Cons

Interoperability³⁴

Reproducibility³

Reusability³

Productivity

Communication

Cost of paying technical debt

Have to update to avoid obsolescence

Lose “benefits” of closed data formats

Standard may not be a perfect fit

Standards may not always be best

- For Eelbrain moving to BIDS
 - Data sharing since open data is often in BIDS format
 - Community has converged on BIDS
 - Start with six BIDS entities, then add more as needed

³Basu et al. (2019)

⁴David and Greenstein (1990)

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Make Backward Compatible:

Reasons to keep	Reasons to break
Existing code still works	Time+effort to keep
Facilitates transition	Future maintenance easier
Reproducibility	Still have tech. debt.
Protects downstream soft.	Removes constraints
User trust	Reduces testing burden

- Depends on current architecture
- For Eelbrain-BIDS, backward compatibility was dropped – BIDS makes different assumptions, too much repetition of code, not enough demand to justify

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- Don't need to understand everything, just enough to understand the interface
- Ask domain expert questions⁵
 - What keyword should I search for?
 - What is the problem that the software is trying to solve?
 - What is the context in which the software is used?
 - What are the typical use cases for the software?
 - **What is the approximate scale of the problem.**
 - What should the user provide as **inputs**?
 - What **output** does the user expect?
 - What are trends in the output as the input varies?
- For a well-documented project, an AI chat-bot could substitute for a domain expert

⁵Du and Smith (2025)

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- Need current architecture for planning
- If the interface is well-defined, change can be isolated
- If not, the first step is to refactor
- Use information hiding principle⁶

⁶Parnas (1972)

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Solve $\frac{dT_W}{dt} = \frac{1}{\tau_W}(T_C - T_W(t))$, $T_W(0) = T_{init}$

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```

1  t = [0.0]
2  T_W = [T_init]
3
4  while t[-1] < t_end:
5      T = T_W[-1]
6
7      # Euler update
8      T_new = T + dt * (T_C - T) / tau_W
9      t_new = t[-1] + dt
10
11     t.append(t_new)
12
13     T_W.append(T_new)

```

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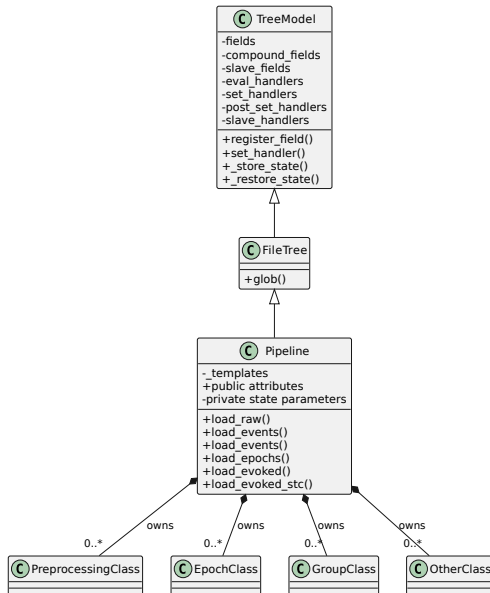
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```

1  def dT_W_dt(t, T_W, T_C, tau_W):
2      return (T_C - T_W) / tau_W
3
4  # Bind the parameters of our particular ODE
5  def derivative(t, T_W):
6      return dT_W_dt(t, T_W, T_C, tau_W)
7
8  t_eval = np.arange(0, t_end + dt, dt)
9
10 solution = solve_ivp(
11     derivative,
12     (0, t_end),
13     [T_init],
14     t_eval=t_eval,
15     method="RK45"
16 )

```



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- Eelbrain-BIDS
 - Changes mostly to Pipeline
 - Missed path templating
- Degree of abstraction depends on likelihood of change
- Wrapper (Adapter) design pattern in trivial cases
- Add strategy pattern for backward compatibility
- Requires a mapping between the old and new interface
- Not an option for Eelbrain-BIDS

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- Which library?
 - Performance, robustness, documentation quality, learning curve, community support, etc.
 - Flexibility vs. simplicity
 - Community size
- **Task decomposition (helps non-experts)**
- Assessing change impact, search code
- Code reading using debug mode
- Incremental iterative development
- Eelbrain-BIDS
 - Upgrade parent classes first
 - Domain knowledge in steps: preprocessing, epochs, source localization, etc.
 - Stick with `mne.read_raw_fiff` over `mne-bids`

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- Multi-layered testing (unit, integ., system)
- Unit tests, integration tests, system tests
- Reuse existing tests
- Continuous integration (CI)
- **Update documentation with each pull request**

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- Version number as clearest signal of breaking changes⁷
- Release notes
- Deprecation warnings
- Evolution models
 - Old and new interfaces coexist for a while, then old interface is deprecated
 - Old interface is deprecated immediately (two in production)⁸
- Eelbrain-BIDS uses two in production

⁷ [Preston-Werner \(2023\)](#)

⁸ [Lübke et al. \(2019\)](#)

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